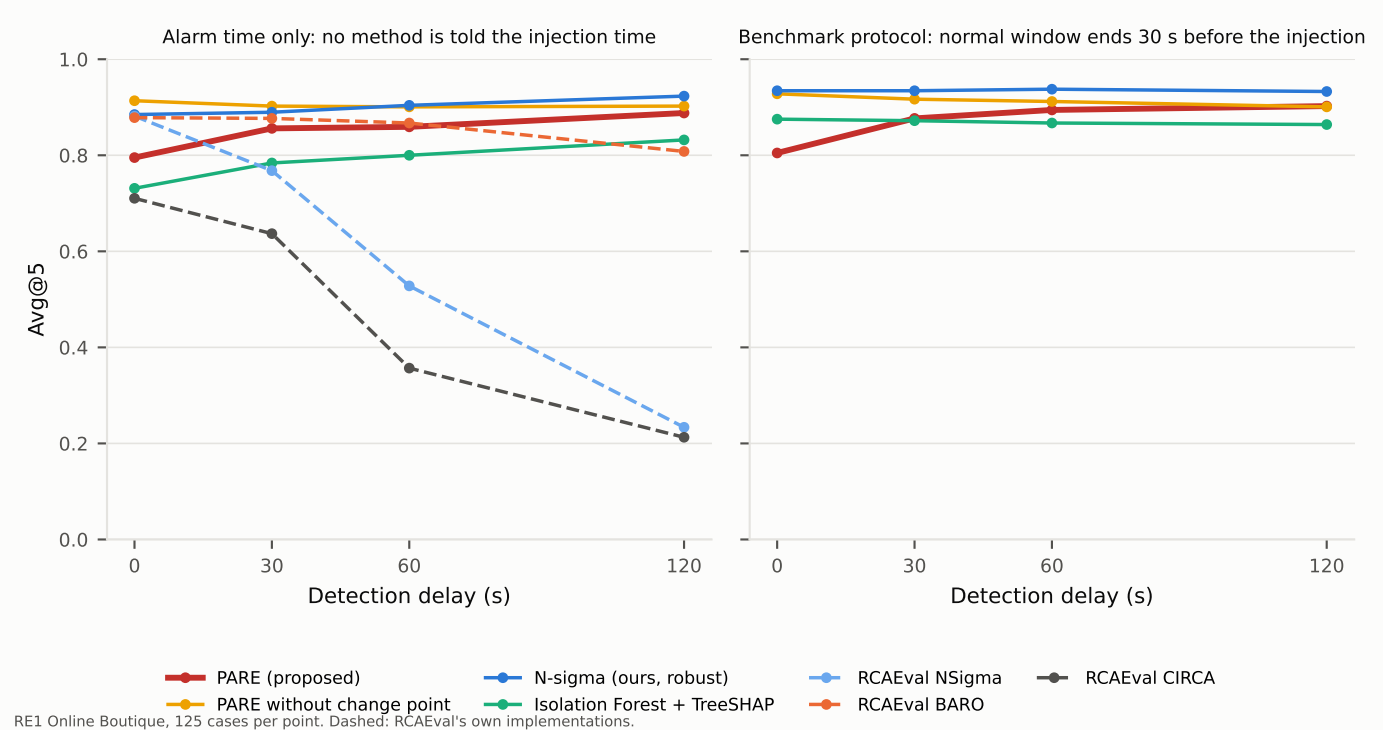
Alarm time only: no method is told the injection time
Benchmark protocol: normal window ends 30 s before the injection
Avg@5
1.0
0.8
0.6
0.4
0.2
0.0
0
30
60
120
Detection delay (s)
PARE (proposed)
PARE without change point
N-sigma (ours, robust)
Isolation Forest + TreeSHAP
RCAEval NSigma
RCAEval BARO
RCAEval CIRCA
RE1 Online Boutique, 125 cases per point. Dashed: RCAEval's own implementations.